

LESSON 6.5

PROPORTIONAL RELATIONSHIPS AND SIMILAR TRIANGLES



LESSON INTRODUCTION

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

LEARNING TARGETS

- I can solve mathematical problems using similar triangles.

BENCHMARK COHERENCE

BUILDING ON

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

WORKING TOWARD

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

ESSENTIAL QUESTIONS

- How can you apply your knowledge of proportional relationships to solve problems using similar triangles?

LESSON NARRATIVE

In this lesson, students apply their knowledge of ratios and proportionality from previous lessons to solve mathematical problems involving similar triangles. Students develop an understanding of similarity and the properties of similar triangles. Then, they solve problems involving similar triangles.

COMPONENT SUMMARY

6.5.1 WARM-UP (~5 MIN.)

Bell Work on finding slope

6.5.3 GUIDED INSTRUCTION (15-20 MIN.)

Discover that corresponding sides of similar triangles are proportional

6.5.5 GUIDED PRACTICE (5 MIN.)

Practice solving problems involving similar triangles

6.5.2 EXPLORATION (10 MIN.)

Explore ratios in similar triangles

6.5.4 COLLABORATION (5 MIN.)

Practice matching similar triangles and finding constants of proportionality

6.5.6 WRAP UP (~5 MIN.)

Reflect on mastery of lesson learning targets

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Similar
- Corresponding sides
- Constant of proportionality

MATERIALS

- Graph Paper
- (Optional) Highlighters
- (Optional) Calculators

ADDITIONAL PREPARATION

- (Optional) Consider creating sorting cards with the triangles in 6.5.4 Collaboration.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly identify corresponding sides.
- Student may not realize that a triangle can be rotated and still be similar.

THINGS TO CONSIDER

- Make sure to place emphasis on the fact that triangles can have different orientations but still be similar. Consider having triangles cut out for students to rotate.
- Consider allowing students to use calculators when finding constant of proportionality.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 6.5.2 Exploration, ask students to initially complete the problems independently, then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 6.5.2 Exploration, students justify their methods and compare their methods to their peers. These discussions and reflections will help build conceptual understanding and procedural fluency.

DIFFERENTIATE INSTRUCTION

Consider creating and displaying an anchor chart with definitions, examples, and visuals.

6.5.1 WARM-UP: FINDING SLOPE

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for small-group instruction on slope. Students use slope again in Unit 7, so providing support before that unit can help students be successful in forthcoming units.



6.5.1 Warm-Up: Finding Slope

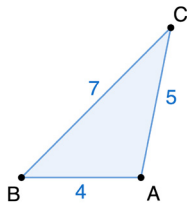
- Find the slope of each line using the given information.

Line	Slope						
	$\frac{\text{Rise}}{\text{Run}}$ $\frac{-4}{2} \text{ or } \frac{-2}{1} \text{ or } -2$						
The line that passes through (4, 1) and (10, 4).	<table border="1"> <tr> <td>x</td><td>y</td></tr> <tr> <td>4</td><td>1</td></tr> <tr> <td>10</td><td>4</td></tr> </table> $\frac{\text{Change in } y}{\text{Change in } x}$ $\frac{3}{6} \text{ or } \frac{1}{2}$ OR $\frac{y_2 - y_1}{x_2 - x_1}$ $\frac{4-1}{10-4} = \frac{3}{6} = \frac{1}{2}$	x	y	4	1	10	4
x	y						
4	1						
10	4						



6.5.2 Exploration: Ratios & Similar Triangles

Triangle *ABC* is shown.



Lauren wants to create three new triangles that have the same shape as triangle *ABC*, but are different sizes. She multiplies each side of triangle *ABC* by the scale factors given in the table to determine the sides of the new triangles.

1. Complete the table to determine the side lengths of the three new triangles, triangle *DFH*, triangle *GKM*, and triangle *SLN*.

Triangle	Scale Factor	Length of Short Side	Length of Medium Side	Length of Long Side
<i>ABC</i>	1	4	5	7
<i>DFH</i>	2	8	10	14
<i>GKM</i>	3	12	15	21
<i>SLN</i>	$\frac{1}{2}$	2	2.5	3.5

2. Complete the table by writing the following ratios.

Triangle	$\frac{\text{long side}}{\text{short side}}$	$\frac{\text{long side}}{\text{medium side}}$	$\frac{\text{medium side}}{\text{short side}}$
<i>ABC</i>	$\frac{7}{4}$ or 1.75	$\frac{7}{5}$ or 1.4	$\frac{5}{4}$ or 1.25
<i>DFH</i>	$\frac{14}{8}$ or 1.75	$\frac{14}{10}$ or 1.4	$\frac{10}{8}$ or 1.25
<i>GKM</i>	$\frac{21}{12}$ or 1.75	$\frac{21}{15}$ or 1.4	$\frac{15}{12}$ or 1.25
<i>SLN</i>	$\frac{3.5}{2}$ or 1.75	$\frac{3.5}{2.5}$ or 1.4	$\frac{2.5}{2}$ or 1.25

3. Compare your answers with your partner. Then, discuss your observations of the ratios.
This is a discussion opportunity. Students do not need to write an explanation here. A sample response is provided
Answers will vary. Possible answers:
When the ratio is divided, the decimals for each column are the same. The numerators and denominators in each column are either factors or multiples of each other. The ratios written as decimals appear to be ordered from greatest to least within the columns moving left to right on the table.

6.5.2 EXPLORATION: RATIOS & SIMILAR TRIANGLES

TEACHER MOVES

Think-Pair-Share

Have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

QUESTIONING

Remind students that ratios can be written as fractions or their equivalent decimal. For example, $\frac{7}{4}$ is equivalent to 1.75.

ENRICHMENT

- Can similar triangles have the same shape but different size?
- Are the corresponding side lengths of similar triangles proportional? How do you know?

6.5.3 GUIDED INSTRUCTION: SIMILAR TRIANGLES AND PROPORTIONALITY

DIFFERENTIATE INSTRUCTION

Students often think that triangles cannot be similar if they have different orientations. Place emphasis on the fact that triangles can have different orientations but still be similar. This can be a hard concept for students to understand because it is visually confusing for them. Consider having triangles cut out for students to rotate.

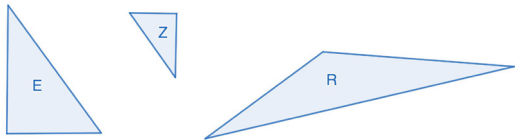
ENRICHMENT

What are the corresponding sides of these triangles? Why is it important to identify corresponding sides first?

6.5.3 Guided Instruction: Similar Triangles and Proportionality

In the Exploration, the triangles created are called **similar** triangles. In mathematics, similar figures have the same shape, but are not necessarily the same size.

- Three triangles are shown.



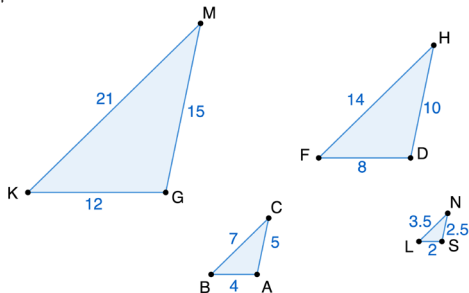
Discuss with your partner which two triangles appear to be similar.

This is a discussion opportunity. Students do not need to write an explanation here. A sample response is provided. The conclusion is that Triangle E and Triangle Z are similar—The angles and side lengths may be noted.

Similarity cannot be determined by appearance alone. Triangles are similar if their corresponding sides are proportional.

The exploration used the words “long,” “medium,” and “short” to describe the sides of the triangles. Sides of similar triangles that are paired are called **corresponding sides**.

Similar triangles ABC , DFH , GKM , and SLN from the exploration are shown.

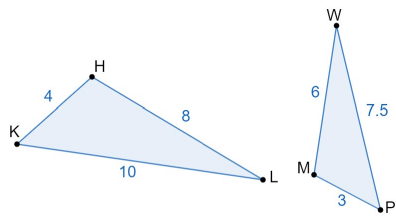


In the exploration, the long side of triangle ABC is the side between points B and C , called segment BC .

- Complete the table by naming the corresponding long sides of the triangles from the exploration.

Triangle	Long Side
Triangle DFH	Segment FH
Triangle GKM	Segment KM
Triangle SLN	Segment LN

3. Triangle *HKL* and triangle *MPW* are similar.



- a. Complete each statement to match the corresponding sides in the two similar triangles.
- Segment *HK* corresponds with segment *MP*.
- Segment *HL* corresponds with segment *MW*.
- Segment *KL* corresponds with segment *PW*.

 In similar triangles, corresponding sides are proportional.

The constant of proportionality between any two similar triangles is the ratio of the corresponding side lengths. For triangle *HKL* and triangle *MPW*, the constant of proportionality is written as $\frac{\text{corresponding side length on } HKL}{\text{corresponding side length on } MPW}$.

6.5.3 GUIDED INSTRUCTION: SIMILAR TRIANGLES AND PROPORTIONALITY (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider printing out the triangles so that students can cut them out if they need to rotate the triangles to compare the sides. This might allow students to match the corresponding sides.

- b. Find the constant of proportionality using each set of corresponding sides from triangle *HKL* and triangle *MPW*.

$\frac{\text{corresponding side length on } HKL}{\text{corresponding side length on } MPW}$	Constant of Proportionality
$\frac{HK}{MP}$	$\frac{4}{3}$ or 1.33 ...
$\frac{HL}{MW}$	$\frac{8}{6}$ or 1.33...
$\frac{KL}{PW}$	$\frac{10}{7.5}$ or 1.33...

- c. With your partner, discuss whether it is necessary to calculate the constant of proportionality for all sets of corresponding sides, or to calculate the constant of proportionality for one set of corresponding sides.
- This is a discussion opportunity.*
Students do not need to write an explanation here.

ENRICHMENT

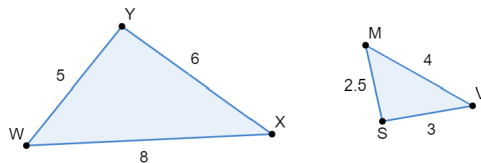
- How can you check if two triangles are similar?
- How can do you find the constant of proportionality between two triangles?
- How does this relate to our previous lessons on the constant of proportionality?

6.5.3 GUIDED INSTRUCTION: SIMILAR TRIANGLES AND PROPORTIONALITY (CONTINUED)

TEACHER MOVES

The mistake that Mansoor makes in problem 4 is a common mistake. Many students struggle with identifying corresponding sides, particularly when the orientation of the triangles is not the same. Identifying and correcting the mistake here may help students avoid making the mistake themselves in the future.

4. Triangle WYX and triangle MSV are shown below.



Mansoor claims that the two triangles are not similar because the ratios of the corresponding sides are not equivalent. His work supporting his claim is shown.

$$\frac{WY}{MS} = \frac{5}{2.5} = 2$$

$$\frac{YX}{MV} = \frac{6}{4} = 1.5$$

- Explain whether you agree with Mansoor. Show any work that supports your answer.
Answers will vary. (Mansoor is incorrect) Mansoor is incorrect because he did not match the correct legs of the triangle. Example: the long leg of one triangle, and the long leg of the other triangle. Or the short leg of one triangle and the short leg of the other triangle.
- Compare your work with your partner and resolve any differences, if necessary.
This is a discussion opportunity. Students do not need to write an explanation here.



6.5.4 Collaboration: Matching Similar Triangles

Eight triangles are shown.

Triangle <i>ABF</i>	Triangle <i>DHY</i>
Triangle <i>GNL</i>	Triangle <i>KMC</i>
Triangle <i>PTR</i>	Triangle <i>NCS</i>
Triangle <i>TWU</i>	Triangle <i>VXY</i>

6.5.4 COLLABORATION: MATCHING SIMILAR TRIANGLES

DIFFERENTIATE INSTRUCTION

Consider having students highlight corresponding sides with the same color to help students identify which sides are corresponding. This will provide students with a visual entry point. Consider creating sorting cards so students can physically match the similar triangles.

QUESTIONING

Consider challenging students who finish early to determine the scale factor for each pair.

ENRICHMENT

- Can you use any pair of corresponding sides to find the constant of proportionality?
- How can you check your work when solving mathematical problems involving similar triangles?

1. Work with your partner to complete the table by determining each pair of similar triangles and finding the constant of proportionality for each pair.

Similar Triangles	Constant of Proportionality
Triangles <i>ABF</i> and <i>GNL</i>	$\frac{4}{3.2} = 1.25$ $\frac{4.5}{3.6} = 1.25$
Triangles <i>NCS</i> and <i>VXY</i>	$\frac{6.5}{3.9} = 1.\bar{6}$ $\frac{5}{3} = 1.\bar{6}$
Triangles <i>KMC</i> and <i>TWU</i>	$\frac{5.5}{2.2} = 2.5$ $\frac{6.5}{2.6} = 2.5$ $\frac{11.5}{4.6} = 2.5$
Triangles <i>DHY</i> and <i>PTR</i>	$\frac{5}{1} = 5$ $\frac{6}{1.2} = 5$ $\frac{9.5}{1.9} = 5$

6.5.5 GUIDED PRACTICE: SOLVING MATHEMATICAL PROBLEMS WITH SIMILAR TRIANGLES

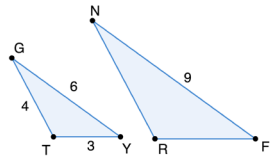
ENRICHMENT

- What is the first step?
- How do you know which sides are corresponding?
- Are the corresponding sides proportional?
- How did you find the constant of proportionality?



6.5.5 Guided Practice: Solving Mathematical Problems with Similar Triangles

1. Triangle GTY is similar to triangle NRF .



- a. What is the constant of proportionality between triangle GTY and triangle NRF ?
 $\frac{9}{6} = 1.5$ or $\frac{6}{9} = 0.\bar{6}$
- b. Find the lengths of the missing sides of triangle NRF .
 $4 \times 1.5 = 6$ $RN = 6$
 $3 \times 1.5 = 4.5$ $RF = 4.5$

- c. Complete the table to verify your work.

corresponding side length on GTY corresponding side length on NRF	Constant of Proportionality
$\frac{GY}{FN}$	$\frac{6}{9} = 0.\bar{6}$
$\frac{GT}{RN}$	$\frac{4}{6} = 0.\bar{6}$
$\frac{TY}{FR}$	$\frac{3}{4.5} = 0.\bar{6}$



6.5.6 Wrap-Up: Reflection

1. For each of the statements below, circle the emoji that represents your feelings toward each statement.

Answers will vary.

- a. I can identify the corresponding sides of similar triangles.



0 1 2 3 4 5 6 7 8 9 10

- b. I can determine whether two triangles are similar.



0 1 2 3 4 5 6 7 8 9 10

- c. I can find the missing side lengths in similar triangles.



0 1 2 3 4 5 6 7 8 9 10

6.5.6 WRAP-UP: REFLECTION

TEACHER MOVES

Consider having students markup text and label each activity with an emoji or draw their own emoji.